

Knowledge-Infused Multimodal Retrieval for Scientific Papers

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Abstract

This paper presents a knowledge-infused multimodal question answering system for scientific research papers, combining structured parsing, retrieval, and evidence attribution.

1. Introduction

Scientific papers contain rich multimodal information including text, tables, and figures. Understanding this information requires a system that can jointly reason over all modalities. In this work we propose a pipeline that parses a PDF into a structured representation before any retrieval or generation takes place.

2. Method

Our method consists of four stages: ingestion, parsing, chunking, and retrieval.

2.1 Parsing

The parsing stage converts a raw PDF into sections, paragraphs, tables, and figures using a document layout model.

Model	Accuracy	F1
Baseline	0.71	0.68
Ours	0.89	0.87

Table 1: Comparison of baseline and proposed method.

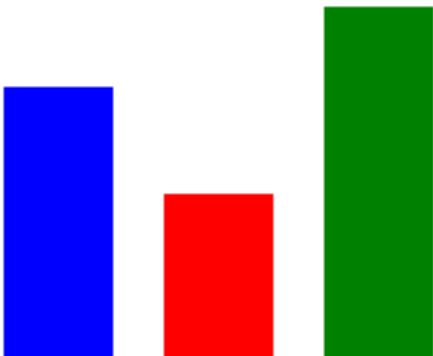


Figure 1: Example bar chart illustrating result magnitudes.

3. Conclusion

We presented a parsing pipeline for scientific documents and showed its role in a larger multimodal question answering system.

References

- [1] Smith, J. (2020). A prior paper on document parsing.
- [2] Doe, A. (2021). Multimodal retrieval methods.